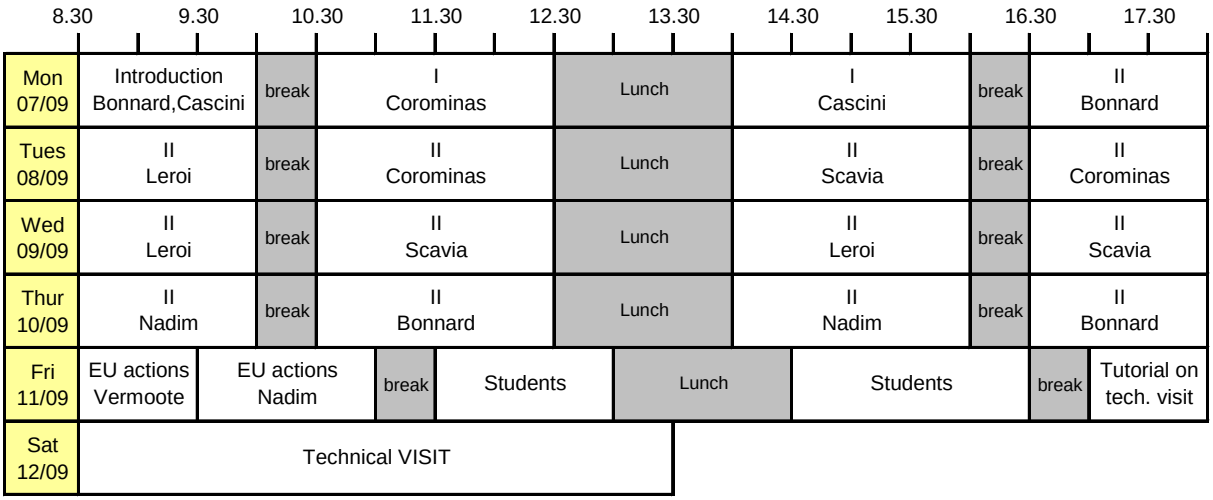
LARAM School 2009: 1st week



INTRODUCTION [1.5 hours]

Bonnard (0.5 hours): Introduction to LARAM 2009
Cascini (1.0 hours): Introduction to landslide risk

SESSION I [4 hours]
"Landslide analysis using different approaches"

Corominas (2.0 hours): Landslides typology and classification. Geological analyses.
Cascini (2.0 hours): The geotechnical slope model.

SESSION II [22.5 hours]
"Risk Theory and Risk Analysis for Landslides"

Bonnard (1.5 hours): Introduction to session II and landslide vulnerability components.
Leroi (1.5 hours): Landslide risk framework and examples.
Corominas (2.0 hours): Assessment of landslide frequency.
Scavia (2.0 hours): Probabilistic approach to the assessment of rock slope stability (part I).
Corominas (1.5 hours): Risk analysis case study.
Leroi (1.5 hours): Landslide susceptibility and hazard.
Scavia (2.0 hours): Probabilistic approach to the assessment of rock slope stability (part II).
Leroi (2.0 hours): Landslide risk management at La Desirade Island.
Scavia (1.5 hours): A simplified GIS-based probabilistic methodology for the evaluation of the susceptibility of rock falls.
Nadim (1.5 hours): QRA theoretical aspects.
Bonnard (2.0 hours): Long-term efficiency of protection works and residual risks.
Nadim (2.0 hours): Deterministic and probabilistic methods for slope stability analysis.
Bonnard (1.5 hours): La Reunion case study tutorial.

EU actions in the field of landslides [2.5 hours]

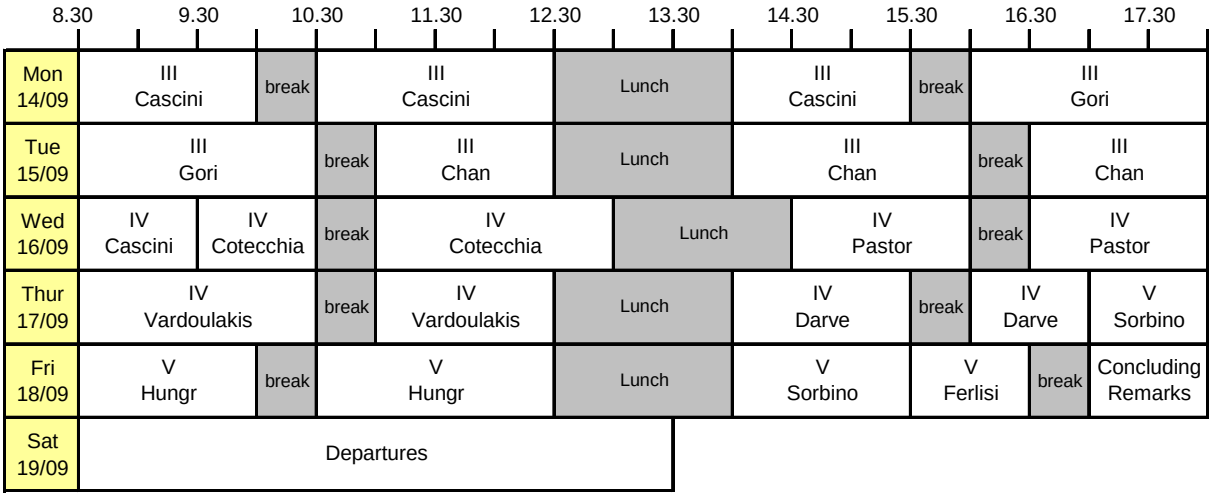
Vermoote (1 hours): EU funded research activities in the field of landslides and the link with current European policy and strategies on natural hazards and disasters.
Nadim (1.5 hours): The "Safeland" FP7 Project.

Presentations by students [3.5 hours]

Tutorial on the Technical Visit [1.0 hours]

Technical Visit [5.0 hours]

LARAM School 2009: 2nd week



Session III [14 hours]
"Landslide susceptibility, hazard and risk zoning: international experiences"

Cascini (5.0 hours): International guidelines for landslide susceptibility, hazard and risk zoning for land use planning. Suggested approaches for analyses at different scales. Tutorial on susceptibility, hazard and risk zoning at 1:25,000 scale. Statistical methods for landslide susceptibility and hazard zoning and Tutorial (Dr. Calvello).

Gori (4.0 hours): Landslide hazard and risk zoning at small scale: the USGS experience.

Chan (5.0 hours): Landslide risk zoning at large scale: the GEO experience.

Session IV [13 hours]
"The role of sophisticated methods in landslide Risk analysis"

Cascini (1.0 hours): Introduction to advanced slope stability characterization.

Cotecchia (3 hours): The geomechanical approach to landslide hazard assessment - interpretation and modelling at different scales.

Pastor (3 hours): Mathematical modelling: triggering and propagation. Rheological modelling: a new constitutive model for unsaturated soils. Numerical modelling: initiation and propagation.

Vardoulakis (3.5 hours): Basic Geodynamics of Landslides. Thermo-poromechanics and grain crushing as origins for the catastrophic acceleration phase of landslides.

Darve (2.5 hours): Failure in geomechanics. Finite element modelling of landslides.

SESSION V [7.0 hours]
"Landslide risk mitigation"

Sorbino (1.0 hours): The role of control works in the risk mitigation framework.

Hungr (3.5 hours): Slope stabilization techniques. Defensive measures against rock fall and rapid landslides.

Sorbino (1.5 hours): Remarks on control works for landslide risk reduction and some case histories.

Ferlisi (1.0 hours): Physical vulnerability to slow-moving landslides.

CONCLUDING REMARKS [1 hour]